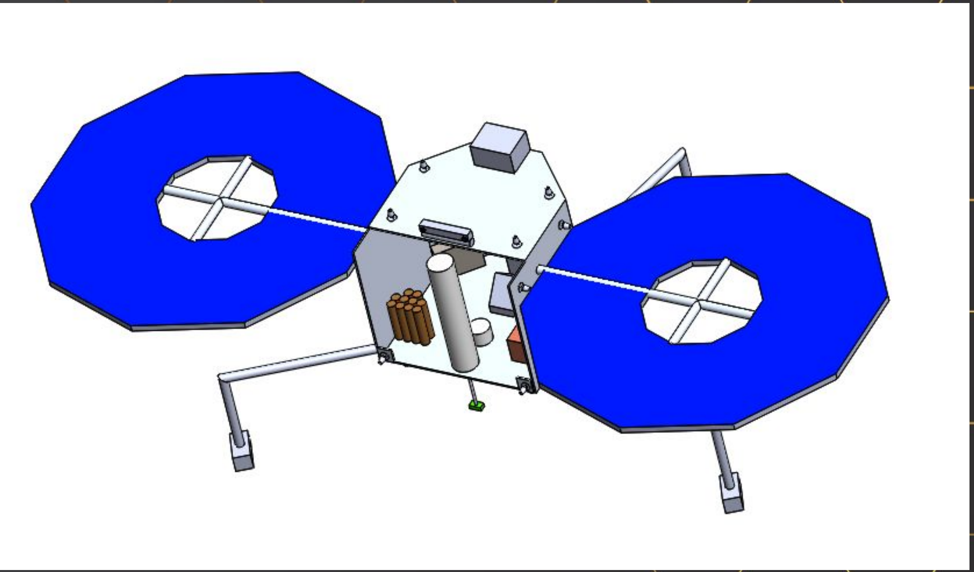
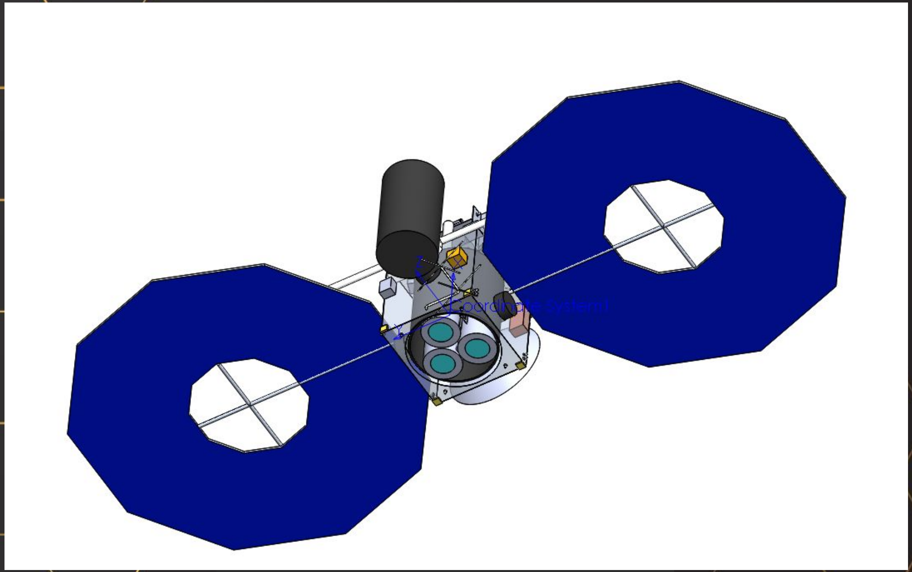
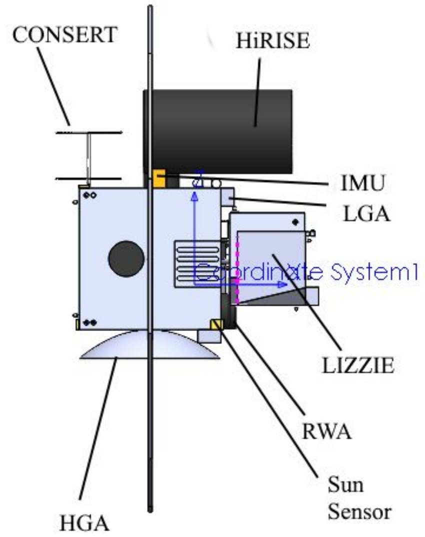


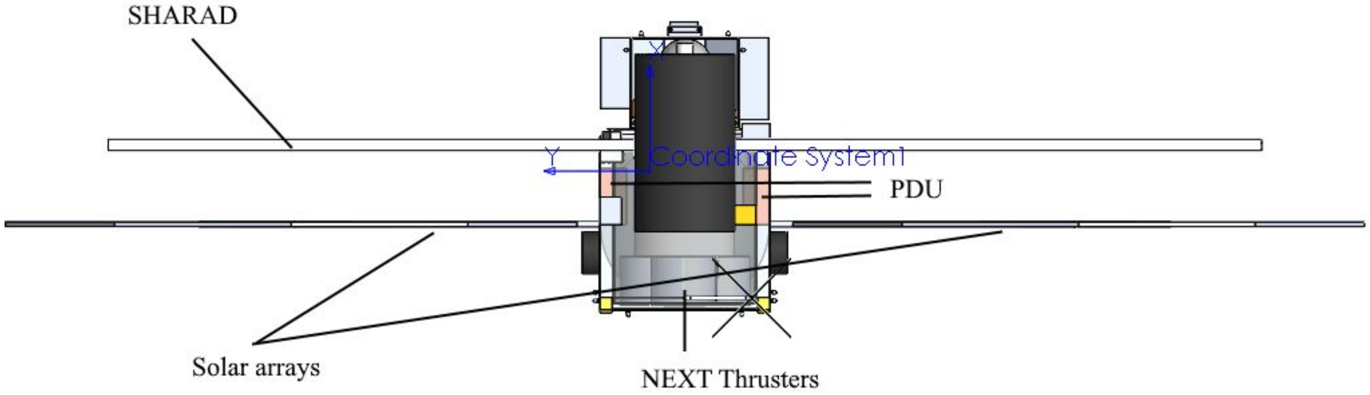


CAD drawing of our spacecraft/lander

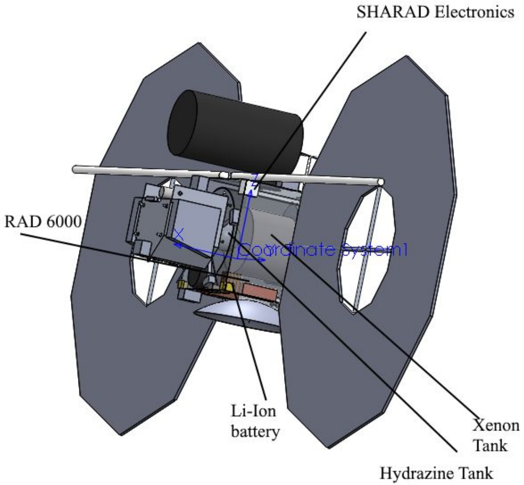
The Spacecraft

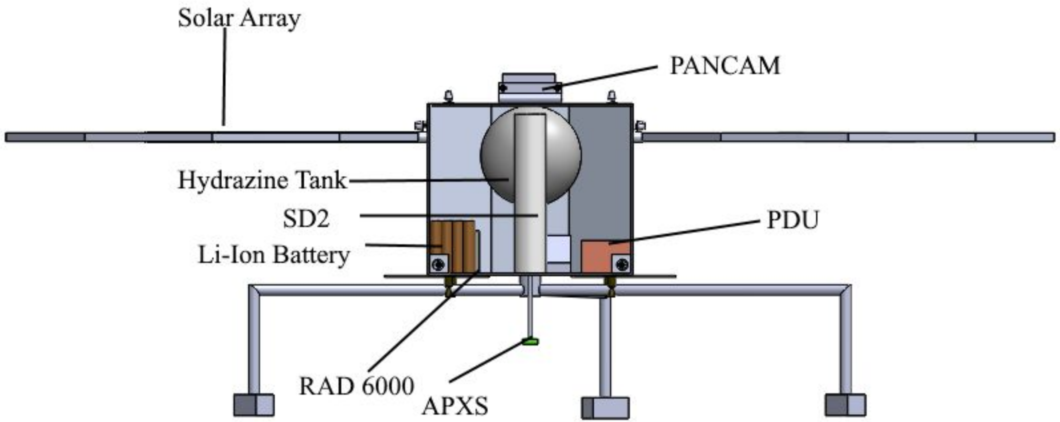
Designed with the goal of maximizing capability and sustainability, our spacecraft was modeled after the Dawn and Rosetta-Philae spacecrafts.

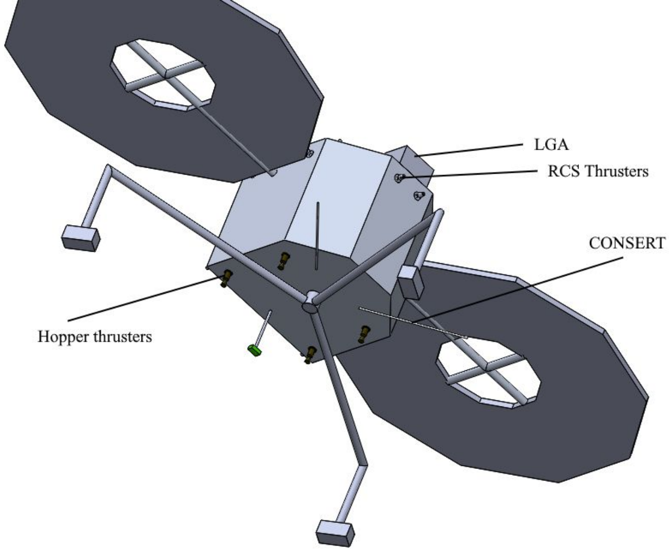




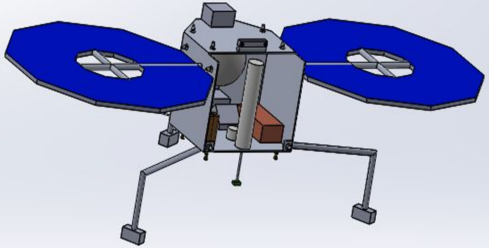
SAMMIE Labelled Views





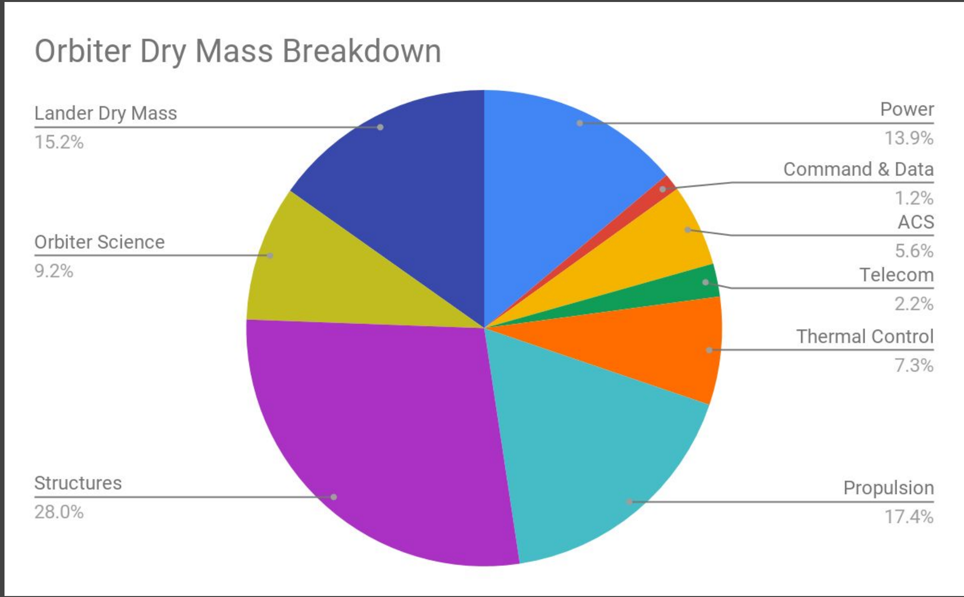


LIZZIE Labelled Views



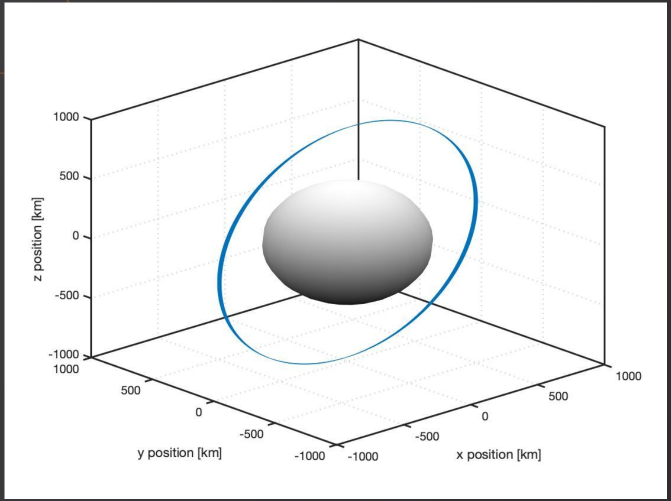
MEL

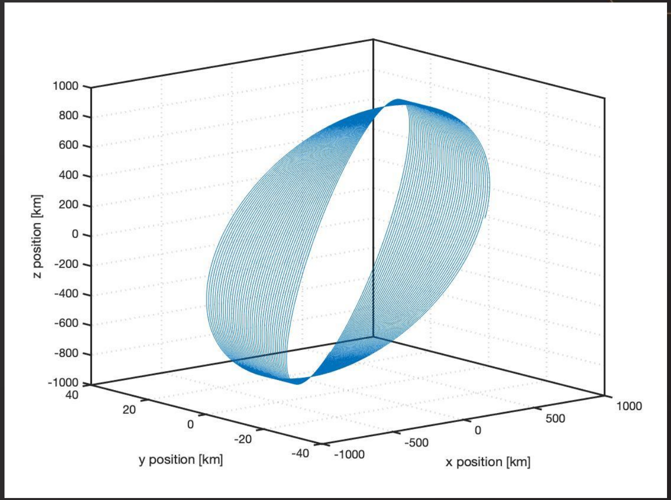
Subsystem/Component	Predicted Mass, kg Total Flight	 Contingency %	 Total Flt w/Contingency
ORBITER BUS	**638.62**	**11.9%**	**714.90**
Power	118.97	10.5%	131.45
Command & Data Handling	10.50	5.0%	11.03
ACS	50.16	5.0%	52.66
Telecom	20.23	5.0%	21.24
Thermal Control	61.60	12.7%	69.42
Propulsion	156.66	5.0%	164.50
Structures	220.50	20.0%	264.60
ORBITER SCIENCE INSTRUMENTS	**83.00**	**5.0%**	**87.15**
ORBITER DRY MASS	**721.62**	**11.1%**	**802.05**
LANDER BUS	**123.02**	**5.0%**	**129.17**
LANDER SCIENCE INSTRUMENTS	**13.98**	**5.00%**	**14.68**
LANDER DRY MASS	**137.00**	**5.00%**	**143.85**
Spacecraft Dry Mass	**858.62**	**10.2%**	**945.90**
EXPENDABLES			
Lander Mass Without Propulsion System			116.72
Lander Hopping/Landing Hydrazine MEV			14.13
		Hydrazine Margin	20%
		Hydrazine Final	16.95
Lander Wet Mass:			**160.80**
RCS Propulsion Hydrazine MEV			29.24
		Hydrazine Margin	20%
		Hydrazine Final	35.09
Spacecraft Neutral Mass:			**997.94**
Spacecraft Mass Without Propulsion System			833.44
		SEP Propulsion Xenon MEV	376.74
		Xenon Margin	20%
		Xenon Final	452.08
Spacecraft Wet Mass at LV Separation:			**1450.0**
Falcon 9 ASDS Capability from CCAFS		LV Capacity (kg):	1700.0
		C3 (km^2/s^2):	12.31
Maximum Possible Resource Value (MPV = LV capacity - propellant mass):			1247.9
Launch Margin ((MPV-neutral mass)/neutral mass):			**25.0%**



Ceres Observation Simulator

Design a Keplerian orbit simulator to assist in optimizing Ceres science orbits





Ceres Mapping Orbit: 500km 90.25° SSO to exploit rotation rate

Map 80% of surface in under 2 weeks

Science Operations Summary

	Mapping Orbits	Post-Landing Orbits	Total	Contingency	Final	Units
Number of Orbits	135.0	232.0	367.0	120%	**807.4**	orbits
Time in Orbit	37.83	46.10	83.9	120%	**184.7**	days
Total Data Downlinked to Earth	302.4	45.7	348.1	120%	**765.9**	Gb
Total Downlink Time	420.1	63.5	483.5	120%	**1063.8**	hr
Max Data per Downlink	2.240	4.571	N/A	120%	**N/A**	Gb
Time Between Downlinks (hr)	3.16	110.4	N/A	120%	**N/A**	hr

PEL

	Mode 1 Power (W) Launch	Mode 2 Power (W) Cruise	Mode 3 Power (W) Cruise (Ceres)	Mode 4 Power (W) Science Operations	Mode 5 Power (W) Telecom	Mode 6 Power (W) Maneuvers	Mode 7 Power (W) Battery Recharge	Mode 8 Power (W) Safe Mode
Payload								
Payload MEV	**5.00**	**5.00**	**5.00**	**127.05**	**5.00**	**5.00**	**5.00**	**5.00**
Payload Margin	30%	30%	30%	30%	30%	30%	30%	30%
Payload Requirements	**6.50**	**6.50**	**6.50**	**165.17**	**6.50**	**6.50**	**6.50**	**6.50**
Spacecraft Bus								
Attitude Control	14.70	65.42	65.42	349.97	349.97	65.42	88.52	88.52
Command & Data Handling	36.75	36.75	36.75	36.75	36.75	36.75	36.75	36.75
Power	0.00	59.43	29.72	29.72	29.72	29.72	29.72	29.72
Propulsion	0.00	2449.65	492.45	51.98	51.98	492.45	51.98	51.98
Structures & Mechanisms	0.00	22.58	22.58	0.00	0.00	22.58	0.00	0.00
Telecom	0.00	0.00	0.00	0.00	13.13	0.00	0.00	0.00
Thermal	1.37	145.33	145.33	145.33	145.33	145.33	158.33	158.33
Bus MEV	52.82	364.15	334.43	613.73	626.86	334.43	365.28	365.28
Bus Margin	30%	30%	30%	30%	30%	30%	30%	30%
Bus Total Requirements	**68.66**	**473.39**	**434.76**	**797.85**	**814.91**	**434.76**	**474.87**	**474.87**
SEP MEV	0.00	2415.00	457.80	0.00	0.00	457.80	0.00	0.00
SEP Margin	10%	10%	10%	10%	10%	10%	10%	10%
SEP Total Requirements	**0.00**	**2656.50**	**503.58**	**0.00**	**0.00**	**503.58**	**0.00**	**0.00**
Total								
Spacecraft Total Requirements	**75.16**	**3136.39**	**944.84**	**963.02**	**821.41**	**944.84**	**481.37**	**481.37**
Total Capacity	**0.00**	**3164.63**	**1418.51**	**1418.51**	**1418.51**	**1418.51**	**1418.51**	**1418.51**